

### FILL-IN-THE-BLANK (10)

Each blank is worth 1 point.

1. A camera captures uncompressed RGB images at a resolution of  $1000 \times 1000$  pixels, with 8 bits per channel.
- (a) How many **megabytes (MB)** are required to store one raw frame? Recall that  $1\text{MB} = 1,000,000$  bytes,  $1\text{byte} = 8$  bits. \_\_\_\_\_ MB are required.

*3MB. File size  $b = M \cdot N \cdot k$ . For 1 channel, we have  $1000 \times 1000 \times 8 \text{ bits} = 1\text{MB}$ ; RGB images have 3 channels, so 3MB.*

- (b) To reduce storage, you may reduce either the spatial resolution or the intensity resolution (bits per channel). If you halve the width and height, what should the new bit-depth per channel be, to achieve the same reduction in storage? \_\_\_\_\_ bits per channel.

*2 bits.  $0.5M \cdot 0.5N \cdot k = M \cdot N \cdot k' \Rightarrow k' = 0.25 \cdot k = 0.25 \cdot 8 = 2$*

2. A drone captures an image of a dark forest at night. 94.1% of the pixels have intensity values uniformly distributed between 10 and 15. The remaining 5.9% of pixels represent bright lights and stars in the sky with an intensity of 255.

- (a) You opt to apply histogram equalization to this image. How many unique intensity values will the image have afterwards? \_\_\_\_\_ unique intensity values.

*7 unique intensity values: {10, 11, 12, 13, 14, 15, 255}*

*Histogram normalization redistributes the **existing discrete intensity levels** according to CDF. It does **not create new intensity levels** beyond those present in the input. While we did not specify the number of pixels in the image, the fact that we mentioned 94% implies that there are at least 50 pixels, making it impossible for quantization to cause a collision, i.e., two values mapping to the same output.*

- (b) Assume fractional values of  $240/255$  and  $15/255$  for the proportion of dark and light pixels, respectively. What would pixels with an original intensity value of 12 map to? 12 maps to \_\_\_\_\_

*12 maps to 120.*

*The dark pixels have a proportion of  $240/255$  as stated. Since the problem states it is uniformly distributed, this means each dark pixel has a proportion of  $40/255$  (divide by the 6 dark values). The cumulative probability of 12 will then be  $120/255$ . This directly maps to a new intensity of 120.*

3. Consider the following  $5 \times 5$  greyscale image of a diagonal edge (1=white, 0=black).

$$I = \begin{bmatrix} 1 & 1 & 1 & 1 & 0 \\ 1 & 1 & 1 & 0 & 0 \\ 1 & 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

Let the notation (r,c) indicate (row, column), with indexing starting at 1. Use Cartesian coordinates (+x points right, +y points upwards) and that angles are measured counterclockwise from the positive x-axis.

Recall the two Sobel kernels

$$S_x = \begin{bmatrix} -1 & 0 & 1 \\ -2 & 0 & 2 \\ -1 & 0 & 1 \end{bmatrix}, \quad S_y = \begin{bmatrix} 1 & 2 & 1 \\ 0 & 0 & 0 \\ -1 & -2 & -1 \end{bmatrix}$$

- (a) You compute image gradients without any border padding. How many unique gradient vectors are present in this image? \_\_\_\_\_ unique gradient vectors.

*6 unique gradient vectors.*

$$G_x = \begin{bmatrix} -1 & -3 & -3 \\ -3 & -2 & -1 \\ -3 & 1 & 0 \end{bmatrix}, \quad G_y = \begin{bmatrix} 1 & 3 & 3 \\ 3 & 2 & -1 \\ 3 & 1 & 0 \end{bmatrix}$$

*The distinct gradient vectors  $(G_x, G_y)$  are:  $\{(-3,3), (-2,2), (-1,1), (-1,-1), (1,1), (0,0)\}$*

*Note, it is not necessary to compute all 9 Sobel responses explicitly; counting the number of distinct 3x3 neighbourhoods in the valid convolution region suffices.*

- (b) What is the gradient magnitude at element (4,4) of the image matrix? Indicate square roots as  $(x)^{0.5}$ . The gradient magnitude is \_\_\_\_\_.

*0. Based on the 3x3 neighbourhood at (4,4), all the values cancel, so the gradient vector is (0,0), leading to a magnitude of 0.*

- (c) What is the gradient orientation at element (3,3)? Indicate in degrees. The orientation is \_\_\_\_\_ degrees.

*135 (or -225). The orientation is defined as  $\tan^{-1} \frac{2}{-2}$ , but one can also see intuitively without doing the computation, that the gradient points towards the upper left corner which corresponds to 135 degrees.*

4. You are building a texture classification system using a texton dictionary and brute-force nearest neighbour-classification based on a sum-of-squared differences.

Your training dataset contains **100 images**, each of spatial size **100 × 100** pixels. A filter bank with **10 filters of size 11x11** is applied to every image. Assume that there is no border padding. The pixel-wise filter responses are clustered using K-means to learn a dictionary of **50 textons**. Each image is represented by a single histogram with one bin per texton.

Fill in the following blanks. If the answer cannot be determined from the information given, put -1.

- (a) The dimensionality of feature vectors fed into the K-means clustering to learn the dictionary is \_\_\_\_\_.

*10. 10 filters, 10 dimensional feature for each pixel.*

- (b) The total number of feature vectors fed into the K-means clustering to learn the dictionary is \_\_\_\_\_.

*810,000. 90x90 (remove the border pixels from the convolution) = 8100 pixels per filtered image; 8,100 x 100 images = 810,000*

- (c) To classify each image, the number of squared-difference terms which must be computed is \_\_\_\_\_.

*5000. 50 squared-difference terms for comparing each image (one per feature dimension, image is represented by 50 bin histogram). This must be done to each of the 100 training images. 50 terms x 100 images = 5000*

### MULTIPLE CHOICE (10)

Each MCQ is worth 1 point.

1. You are building an algorithm to detect a specific orange basketball in a video feed. The lighting conditions vary significantly (e.g., bright sunlight vs. shadows), which changes the pixel intensity. Which colour representation would be most robust for defining the "orange" colour independent of brightness?
- Raw RGB values
  - Greyscale intensity, based on RGB weights  
 $\{w_R = 0.8, w_G = 0.2, w_B = 0\}$
  - Greyscale intensity, based on RGB weights  
 $\{w_R = 0.299, w_G = 0.587, w_B = 0.114\}$
  - Normalized RGB**
  - The 'V' channel in HSV

*Normalized RGB separates colour from intensity. It removes the shading ( $r+g+b=1$ ), so an object's normalized colour remains consistent even if the illumination intensity changes. Raw RGB mixes intensity and colour.*

2. Recall the image formation model  $f(x, y) = i(x, y) \cdot r(x, y)$ . A flat, uniform piece of white paper lies on a desk and is illuminated by a desk lamp positioned to the left. As you scan the pixels from the left side of the paper to the right side, how do the model components change?
- $r(x, y)$  decreases;  $i(x, y)$  remains constant, so
  - $r(x, y)$  remains constant;  $i(x, y)$  decreases**
  - $r(x, y)$  increases;  $i(x, y)$  decreases
  - $r(x, y)$  decreases;  $i(x, y)$  increases
  - $r(x, y)$  remains constant;  $i(x, y)$  remains constant

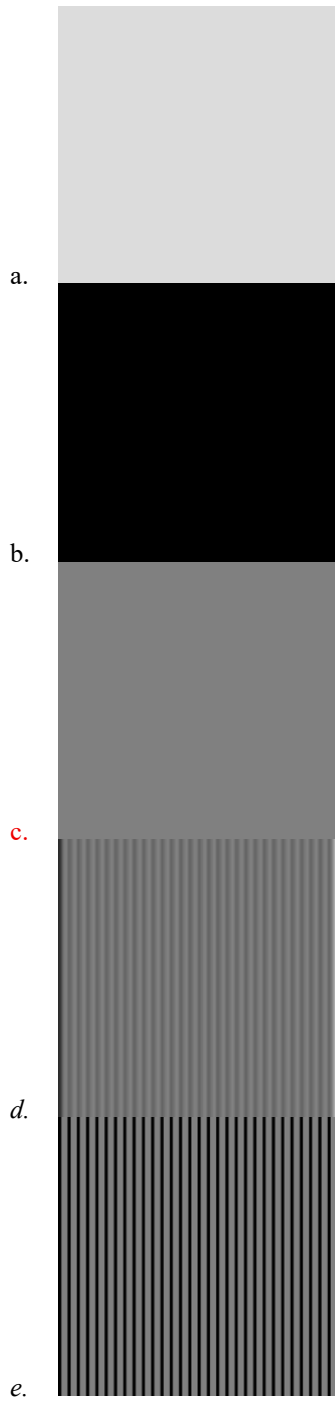
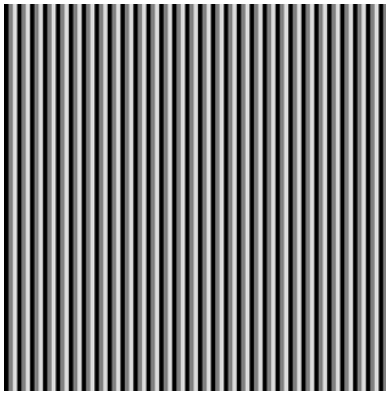
*Of the two components,  $r(x, y)$  is the reflectance, which is a property of the material (uniform white paper), so it is constant.  $i(x, y)$  is the illumination, which falls off as you move further from the lamp.*

3. An image has a minimum intensity of 100 and maximum of 200. After applying a point processing operation, some resulting pixel values become negative. Which operation was definitively applied?
- Histogram stretching
  - Image normalization**
  - Gamma correction with  $\gamma < 1$
  - Histogram equalization
  - Median filtering

*Image normalization (whitening) scales and offsets pixel values so that they are zero-mean and have unit variance. Because the mean is shifted to 0, any original pixel values below the image mean will naturally become negative.*

*Histogram stretching maps the minimum value to 0. Gamma correction is a power law function; inputs greater than 0 will always lead to positive outputs. Median filtering only replaces values with others in the image, so actually none of the pixels will be less than 100. Note that the minimum and maximum intensities, although provided, are not relevant.*

4. Consider the following image with alternating white, grey, and black stripes, each exactly 1 pixel wide. You apply a  $3 \times 3$  median filter to this image. Which image will be the resulting image if you do not apply any border padding?



*Since there are more grey values, it will always change both the black and the white values into grey values. (d) is Gaussian smoothing*

5. Consider line detection in a noisy edge map with spurious edge pixels. Why is the Hough Transform more suitable over least-squares line fitting that minimizes the average squared distance?
  - a. Least-squares line fitting cannot represent vertical lines because it uses a  $y = mx + b$  parameterization, whereas the Hough Transform uses a normal form that can represent all orientations.
  - b. Least-squares line fitting is computationally more expensive than the Hough Transform for large images.
  - c. *Least-squares line fitting minimizes a global squared error and is sensitive to outliers, whereas the Hough Transform accumulates local votes in parameter space, making it more robust to scattered noise.*
  - d. Least-squares line fitting requires the number of lines to be known in advance, while the Hough Transform does not.
  - e. Least-squares assumes Gaussian noise on the dependent variable  $y$ ; this assumption is violated by binary edge maps.

*Squaring the distance term in least squares heavily penalizes points far from the line. If there are many noisy edge points (outliers), they will heavily drag the least-squares line away from the true edge. The Hough transform avoids this because outliers cast random, non-intersecting votes in the parameter space, allowing the true line to emerge as a distinct local maximum.*

6. Recall the normal parameterization of a line, where  $\rho = x \cos \theta + y \sin \theta$ . Which of the following parameter pairs  $(\rho_1, \theta_1), (\rho_2, \theta_2)$  represents an image with two straight lines perpendicular to each other?
  - a.  $\theta_2 = \theta_1$
  - b.  $\theta_2 = \theta_1 + \frac{\pi}{2}$
  - c.  $\theta_2 = \theta_1 + \pi, \rho_2 = -\rho_1$
  - d.  $\theta_2 = \theta_1 + \frac{\pi}{4}, \rho_2 = \rho_1$
  - e.  $\theta_2 = \theta_1 + \pi, \rho_2 \neq \rho_1$

*Two lines are perpendicular if their normals  $\theta$  differ by  $\frac{\pi}{2} \pmod{\pi}$ . For the other options:  $\theta_2 = \theta_1$  (two parallel lines, possibly 1 line without knowing  $\rho$ );  $\theta_2 = \theta_1 + \pi, \rho_2 = -\rho_1$  (represents the same line),  $\theta_2 = \theta_1 + \frac{\pi}{4}, \rho_2 = \rho_1$  (45 degree line),  $\theta_2 = \theta_1 + \pi, \rho_2 \neq \rho_1$  (two parallel lines).*

7. In the standard circle Hough Transform, where the radius is unknown, the vote accumulator is 3D:  $(a, b, r)$ . If the gradient orientation  $\phi$  at an edge point  $(x_i, y_i)$  is used to constrain the possible circle centres, what geometric shape would the point's votes form in the 3D accumulator space?
  - a. One 3D cone
  - b. One 2D circle
  - c. Two 2D circles that intersect each other
  - d. Two discrete points
  - e. *2 straight lines projecting through the 3D space*

*For any given radius  $r$ , the gradient constrains the centre  $(a, b)$  to exactly two points. Consider only one of these two points, in this case, it would be  $a = x_i - r \cos \phi, b = y_i - r \sin \phi$ . As  $\phi, x_i, y_i$  are all fixed,  $a, b$  would change linearly with respect to  $r$ , forming a straight line through the 3D space. For the second point, we have  $a = x_i + r \cos \phi, b = y_i + r \sin \phi$ , which also forms a straight line.*

8. You apply the standard Generalized Hough Transform for an irregular object. During offline modelling, you construct a table that stores displacement vectors from edge points to a chosen reference point, indexed by the gradient orientation (this table is commonly called the R-table). Which of the following modified object-model combinations would still result in the correct detection?
  - a. *Line segment, edge point*
  - b. *Line segment, line segment*
  - c. *Line segment, circle*
  - d. *Circle, edge point*
  - e. *Circle, line segment*

- a. The object is shifted by  $(\Delta x, \Delta y)$ ; before detection, all stored displacement vectors in the R-table are also offset by  $(\Delta x, \Delta y)$ .
- b. The object in the scene is uniformly scaled by a factor of 2; before detection, all stored displacement vectors in the R-table are multiplied by 2.
- c. The object in the scene is rotated by  $30^\circ$  clockwise around the reference point. Before detection, the gradient orientations in the R-table are adjusted by subtracting  $30^\circ$ .
- d. The object is mirrored horizontally. Before detection, the displacement vectors are also mirrored horizontally.
- e. The object is rotated by  $30^\circ$  clockwise around the reference point. Before detection, the stored displacement vectors in the R-table are rotated by  $30^\circ$  clockwise.

*The object is shifted by  $(\Delta x, \Delta y)$ . The Generalized Hough Transform is designed to handle translation. No need to do anything, so offsetting the displacement vectors in the R-table will actually lead to the wrong result. If the object in the scene is rotated by  $30^\circ$  counter-clockwise (or mirrored horizontally), we need to adjust the gradient orientation and the displacement vectors accordingly.*

9. In lectures, the mean-shift algorithm was illustrated using a radially symmetric window with a single bandwidth parameter  $h$ , so the same window extent was applied uniformly across all feature dimensions. In practice, the formulation is extended to multiple bandwidth parameters, to allow for different scaling of the feature dimensions.

You apply mean-shift clustering in a joint spatial-colour feature space using bandwidths  $h_s$  (spatial) and  $h_c$  (colour). The algorithm converges, but the resulting segmentation produces hundreds of small, spatially contiguous regions that differ only slightly in colour. Which adjustment is most appropriate to reduce the number of resulting segments?

- a. Increase  $h_s$ ; decrease  $h_c$ .
- b. Decrease  $h_s$ ; increase  $h_c$ .
- c. Keep  $h_s$  fixed; increase  $h_c$ .
- d. Increase  $h_s$  fixed; keep  $h_c$  fixed.
- e. Decrease  $h_s$ ; decrease  $h_c$ .

*The fragmentation is driven by small colour differences, creating many modes in colour space  $\rightarrow h_c$  is too small. Increasing  $h_c$  enlarges the neighbourhood in the colour dimensions, merging nearby colour modes while preserving the spatial structure. Increasing  $h_s$  may further reduce the number of segments, but it is stated that the regions are already spatially contiguous, so this is not appropriate.*

10. A student applies standard K-means clustering in a 5D feature space of  $(R, G, B, x, y)$  to segment an image into 100 regions. The result has irregular boundaries and the algorithm converges slowly. They switch to SLIC superpixels instead. What key limitation of standard K-means does SLIC address?

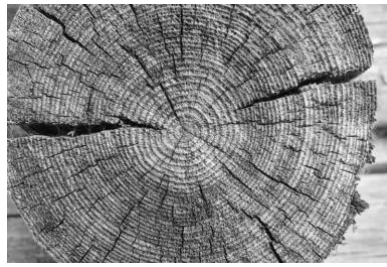
- a. Standard K-means treats the spatial and colour dimensions uniformly; SLIC modifies the distance metric to balance spatial and colour components differently.
- b. Standard K-means evaluates distances from every pixel to every cluster centre globally; SLIC restricts distance computations to a local  $2s \times 2s$  region around each centre.
- c. Standard K-means uses Euclidean distance; SLIC replaces it with a perceptually uniform metric in alternate colour spaces.
- d. Standard K-means optimizes a non-convex objective; SLIC reformulates the problem as a convex optimization.
- e. Standard K-means updates cluster centres using the mean; SLIC updates them using the median to improve robustness.

*SLIC's key improvement is localized search to reduce complexity and promotes spatial compactness. While SLIC weighs spatial and colour components differently, this is not the fundamental limitation addressed in the problem description. Similarly, changing the colour space will not fix the boundaries nor slow convergence caused by the global cluster. Both K-means and SLIC optimize the same non-convex objective. This is a distractor answer. SLIC also does not use median-based updates.*

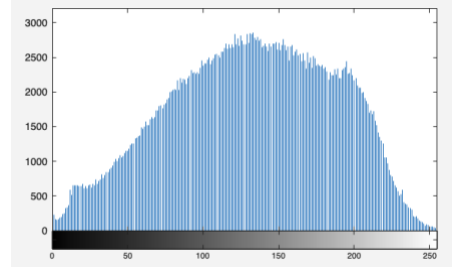
### System Design (6 points)



(a) coloured image of tree cross-section



(b) greyscale image of tree cross-section



(c) Intensity histogram of the greyscale image

Consider the image of a cross-section of a tree in (a). It shows growth “rings”, or concentric bands, where each band corresponds to one year of growth. Your task is to count the number of rings (to estimate the tree’s age) and estimate the relative width of the bands (to infer climate conditions for each year).

Begin by converting the image into greyscale. Then, enhance the image to make the rings more prominent. Finally, design a strategy to estimate the number of rings and width of each ring.

#### Q1: Greyscale conversion (1 point)

If the RGB image has pixel intensities  $\{R, G, B\} \in [0,1]$  and the greyscale intensity  $I$  is obtained by the linear combination  $I = w_R R + w_G G + w_B B$ , where  $\{w_R, w_G, w_B\}$  are scalar weights. Which of the following are feasible values for  $\{w_R, w_G, w_B\}$ ? Select all that apply.

~~$\{0.33, 0.33, 0.33\}$~~ ,  $\{0.5, 0.5, 0.5\}$ ,  $\{0.2, 0.7, 0.1\}$ ,  $\{-0.2, 1.0, 0.2\}$ ,  $\{0.4, 0.4, 0.2\}$ ,  $\{0.1, 0.2, 0.3\}$

*No negative weights, must sum up to 1.*

#### Q2: Ring enhancement (1 point)

You want to process the greyscale image so that the darker bands in the rings are more prominent. The image and its corresponding histogram is shown in (b) and (c) respectively. Which of the following operations are suitable? Select all that apply.

Histogram equalization.

Histogram scaling.

Histogram normalization.

~~Intensity scaling, with  $\alpha > 0$ .~~

Gamma correction, with  $\gamma < 1$ .

Box filtering.

Median filtering with a  $3 \times 3$  kernel.

Normalized cross-correlation with a circular template.

*Histogram equalization and intensity scaling with  $\alpha > 1$  would both increase image contrast. In the latter case, for gamma correction, this relies on observing that the rings are darker tones and using a  $\gamma < 1$  lift these mid-levels to be brighter.*

*We disregarded the solution on intensity scaling as it should be  $\alpha > 1$ , not 0.*

#### Q3: Ring Counting and Width Estimation (4 points)

Assume that you start with an image in which the rings are already sufficiently enhanced from pre-processing. Detail a method for estimating the number of rings and the width of each ring; note the rings are not perfectly circular. State key assumptions for your solution and outline potential weaknesses.

*Rubric: 0.5 points assumption, 0.5 points weakness, 2 points counting method, 1 point width;*

*Note this is just one potential solution – as long as you wrote something reasonable we awarded marks.*

*Solution:*

*This solution exploits the radial structure of the rings.*

*Take multiple radial scanlines extending outward from the estimated center of the tree cross-section. For each scanline, compute the intensity profile along the line. The darker bands of the rings appear as troughs in the intensity profile (or equivalently peaks in the gradient magnitude).*

*Counting the number of troughs along a scanline yields the number of rings intersected by that scanline. To estimate the total number of rings, repeat this procedure over many radial directions and average the counts across scanlines. To ensure that only complete rings are counted, discard scanlines whose profiles terminate prematurely at the image boundary or intersect incomplete rings near the edge of the image. (2 points)*

*The width of each ring can be estimated by measuring the distance between successive troughs (i.e., successive ring boundaries) along each scanline. Because the rings are not perfectly circular, the spacing between troughs will vary with angle. Therefore, compute the width for each radial direction and average the distances across multiple scanlines to obtain a robust estimate of the width of each ring. (1 point for width estimation)*

*No points for generic answers like “solve with deep learning”.*

*Assumption:*

*\*\* Key assumption: the center and edge of the tree cross-section can be located reliably. Secondary assumption: the ring boundaries correspond to detectable intensity transitions. While this is important, it is somewhat already hinted when we mention that the rings are sufficiently enhanced.*

*We can give a stand-alone assumption, or can elaborate how to do this. Detect outer boundary via filtering / segmentation; center can be found based on region centroid, fitting an approximate circle or ellipse to the outermost contour.*

*Award full (0.5) points as long as the most critical assumptions of the solution is given.*

*No points if you assume the tree is a perfect circle, since we already specified this is not the case in the question.*

*Weaknesses:*

*Cracks, knots and defects produce false edges.*

*The width of the ring is not consistent across the entire ring, so averaging across the scanlines is noisy.*

*Error in the center estimation can distort radial measurements. This can be somewhat rectified by adjusting the scan-line to be orthogonal to the intensity gradients*

*Award 0.5 points as long as some weakness is given.*